

Zander Carpio

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Mathematics undergraduate working on quantitative and analytical problems: epidemic dynamics, geospatial disease surveillance, official statistics, and computational number theory. Seeking analytical, data and research roles.

Education

BS Mathematics Aug 2023 to 2027 (expected)
University of the Philippines Baguio
DOST-SEI Undergraduate Scholar, a competitive national merit scholarship from the Philippine Department of Science and Technology. Undergraduate thesis in number theory.
Relevant coursework: Real Analysis, Complex Analysis, Analytic Number Theory, Operations Research, Physics, Accounting.

Experience

Intern, Statistical Operations and Coordination Division Jun to Jul 2026
Philippine Statistics Authority, Baguio City

- Statistical operations, data visualisation and coordination work inside the national statistics agency responsible for official statistics, the census and civil registration.
- Worked with survey and administrative data in its delivered state, including inconsistent labelling and failed joins.

Peer Mentor, Learning Resource Center Sep 2025 to present
University of the Philippines Baguio, Contract
Mentoring undergraduates through mathematics coursework.

Mathematics Tutor Jul 2021 to Jul 2024
Freelance, Hybrid
Three years tutoring mathematics at secondary and lower undergraduate level.

Social Media Manager Jul 2022 to present
Freelance, Remote
Four years running social media for small businesses alongside a full-time degree. Campaign management, scheduling and reporting in GoHighLevel and SocialPilot, with graphic design and content produced in house. Client names withheld.

Quantitative Projects

Dengue surveillance analysis, Baguio City R, QGIS Jan 2024
Reconstructed barangay-level surveillance across 127 barangays and 23 reporting weeks, joined on PSGC codes and mapped as a choropleth. Showed the burden is concentrated rather than citywide, the worst tenth of barangays carrying 32% of cases, and confirmed this was not an artefact of barangay size by re-deriving it from population-normalised incidence (32.4% against 31.7%). That check exposed a denominator failure in the commercial core, where residential population understates the at-risk daytime population enough to yield nominal rates above 4,000 per 1,000.

Compartmental epidemic models R, deSolve, Shiny May 2026
SIR and SEIR systems with mass-action incidence, integrated with `deSolve` and published as interactive Shiny applications in which the parameters can be varied and the epidemic curve re-solved.

Cholera under dual transmission, vaccination and disinfection MATLAB, R May 2026
Four-author paper with A. G. C. Alilio, J. M. Anggaco and K. G. Canor. Extended a published SEIRB model with an exposed compartment for the incubation delay, proved well-posedness, derived both equilibria and obtained R_0 by the next generation matrix. The disease-free equilibrium is locally asymptotically stable for $R_0 < 1$. The structural change moved the policy conclusion: vaccination became the dominant lever on R_0 where the basis model had emphasised environmental control.

Kaprekar triples, undergraduate thesis PARI/GP, LaTeX 2026, ongoing
Extending Iannucci and Foster (2005). Reduced an infinite search to a finite one by proving every triple is a root of a congruence, bounding the count at four per power of ten. Worked to a rule that no result counts until two independently written programs agree, which exposed a counting error in earlier code.

Technical Skills

Languages	R, Python, MATLAB, C, PARI/GP
Libraries	deSolve, Shiny, ggplot2, dplyr, tidyr, NumPy
Tools	QGIS, LaTeX, Git, Excel, GoHighLevel, SocialPilot